Quantum Chemistry Project Report: Benchmarking Numerical Methods for Electron Repulsion Integrals

Mohammad Ahmad: 2074000 Supervisor: Dr. Johannes Margraf

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Abstract

Electron repulsion integrals (ERIs) are the primary computational bottleneck in quantum chemistry simulations, scaling up to $\mathcal{O}(N^4)$ in conventional analytical approaches. Hence, it makes it difficult to run simulations that aid in the development of novel materials and chemicals for various applications. This report presents a systematic benchmark of seven numerical methods for computing ERIs, ranging from brute-force direct summation in real space to uniform Fast Fourier Transforms (FFT), Non-Uniform Fast Fourier Transform (NUFFT), and a two-step Poisson-NUFFT solver. Each method on the hydrogen molecule H_2 is evaluated using the `def2-svp` basis set, measuring absolute error against the analytical PySCF reference, wall-clock time, and peak memory consumption as the grid resolution N is raised. Our results demonstrate that NUFFT methods with a spherical k-space grid achieve the fastest convergence, reaching errors below 10^{-3} at moderate grid sizes, while maintaining competitive computational cost. Despite both two-step Poisson and one-step NUFFT showing similar convergence, the Poisson equation is generally more stable, but also at a slight cost of an increase in time complexity.

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1 Introduction

Many advances in materials science, battery technology, semiconductor design, and superconductivity owe their progress to quantum chemistry simulations. Fundamentally, these simulations approximate solutions to the many-body Schrödinger equation:

$$\hat{H}\,\Psi(\mathbf{r}_1,\ldots,\mathbf{r}_n) = E\,\Psi(\mathbf{r}_1,\ldots,\mathbf{r}_n),$$

where $\hat{H}$ is the molecular Hamiltonian and Ψ is the many-electron wavefunction. Under the Born–Oppenheimer approximation, the nuclei are treated as stationary point charges and the electronic Hamiltonian becomes:

$$\hat{H}_{\text{elec}} = -\frac{1}{2}\sum_i \nabla_i^2 - \sum_{i,A}\frac{Z_A}{|\mathbf{r}_i-\mathbf{R}_A|} + \sum_{i<j}\frac{1}{|\mathbf{r}_i-\mathbf{r}_j|},$$

where the three terms represent the kinetic energy, nuclear-electron attraction, and electron-electron repulsion, respectively. The final term is the most computationally demanding.

To discretise the problem, the molecular orbitals expand in a basis of K atom-centred functions $\{\chi_\mu\}$:

$$\phi_i(\mathbf{r}) = \sum_{\mu=1}^{K} C_\mu\,\chi_\mu(\mathbf{r}). \tag{1}$$

Gaussian-Type Orbitals (GTOs) are the standard choice because the product of two Gaussians centred at different points yields another Gaussian (the Gaussian Product Theorem), which simplifies integral evaluation:

$$\chi_\mu(\mathbf{r}) = N_\mu\,(x-A_x)^{l_x}(y-A_y)^{l_y}(z-A_z)^{l_z}\,\exp\!\big(-\alpha_\mu|\mathbf{r}-\mathbf{A}|^2\big).$$

Substituting (1) into the energy expression produces four classes of molecular integrals. The overlap integrals $S_{\mu\nu}$, kinetic energy integrals $T_{\mu\nu}$, nuclear attraction integrals $V_{\mu\nu}$, and the four-centre electron repulsion integrals (ERIs):

$$(\mu\nu|\lambda\sigma) = \iint \frac{\chi_\mu(\mathbf{r}_1)\chi_\nu(\mathbf{r}_1)\;\chi_\lambda(\mathbf{r}_2)\chi_\sigma(\mathbf{r}_2)}{|\mathbf{r}_1-\mathbf{r}_2|}\,d^3r_1\,d^3r_2. \tag{2}$$

For a basis of size K, the number of unique ERIs scales as $\mathcal{O}(K^4)$, making their evaluation the dominant computational bottleneck. A key insight is that the Coulomb kernel has a simple form in Fourier (reciprocal) space. The Fourier transform of $1/|\mathbf{r}|$ in three dimensions is:

$$\mathcal{F}\left[\frac{1}{|\mathbf{r}|}\right](\mathbf{k}) = \frac{4\pi}{|\mathbf{k}|^2}. \tag{3}$$

This equation transforms the six-dimensional integral in (2) into a product of three-dimensional Fourier transforms, reducing it by the convolution theorem to:

$$(\mu\nu|\lambda\sigma) = \frac{1}{(2\pi)^3}\int \frac{4\pi}{|\mathbf{k}|^2}\;\widetilde{f}(\mathbf{k})\;\widetilde{g}^*(\mathbf{k})\,d^3k,$$

where $\widetilde{f}(\mathbf{k})$ and $\widetilde{g}(\mathbf{k})$ are the Fourier transforms of the overlap densities $f(\mathbf{r}) = \chi_\mu(\mathbf{r})\chi_\nu(\mathbf{r})$ and $g(\mathbf{r}) = \chi_\lambda(\mathbf{r})\chi_\sigma(\mathbf{r})$.

The Coulomb potential generated by a charge distribution $\rho(\mathbf{r})$ satisfies Poisson's equation:

$$\nabla^2 V(\mathbf{r}_1) = -4\pi\,\rho(\mathbf{r}_1), \tag{4}$$

with the solution:

$$V(\mathbf{r}_1) = \int \frac{\rho(\mathbf{r}_2)}{|\mathbf{r}_1 - \mathbf{r}_2|}\, d^3 r_2.$$

Fourier-transforming (3) converts the differential equation into the algebraic relation:

$$\widetilde{V}(\mathbf{k}) = \frac{4\pi}{|\mathbf{k}|^2}\,\widetilde{\rho}(\mathbf{k}),$$

which replaces a costly real-space integration with a pointwise multiplication in k-space, is the foundation of the two-step Poisson–NUFFT solver benchmarked in this study.

The standard FFT operates on uniformly spaced grids and achieves $\mathcal{O}(N \log N)$ complexity [1]. However, molecular systems are better described by non-uniform grids that concentrate points near the nuclei. The Non-Uniform Fast Fourier Transform (NUFFT) generalises the FFT to handle arbitrary point distributions. It computes sums of the form:

$$F(\mathbf{k}_j) = \sum_{m=1}^{M} c_m\, e^{-i\,\mathbf{k}_j \cdot \mathbf{x}_m},$$

where $\{\mathbf{x}_m\}$ are non-uniform source points and $\{\mathbf{k}_j\}$ are non-uniform target points (this is the type-3 NUFFT). The algorithm achieves $\mathcal{O}(M \log M + N \log N)$ complexity via interpolation onto an auxiliary uniform grid, FFT, and deconvolution [2].

The goal of this study is to benchmark several numerical approaches for computing ERIs, listed here: on Cartesian and Becke grids in real space; a Poisson solver on a Cartesian grid; and, finally, on Cartesian and spherical k-space grids. We evaluate convergence behaviour, computational cost, and memory usage, and then, based on the results, analyze the mathematical reasons behind the observed performance differences.

2 Theoretical Background

To evaluate integrals numerically, we replace the continuous integral with a weighted sum over grid points:

$$\int f(\mathbf{r})\, d^3 r \approx \sum_{i=1}^{N} w_i\, f(\mathbf{r}_i),$$

where the choice of points $\{\mathbf{r}_i\}$ and weights $\{w_i\}$ determines both accuracy and efficiency.

The simplest approach is to tile the cubic domain $[-L, L]^3$ with N^3 equally spaced points. The spacing is $\Delta x = 2L/(N-1)$ and each weight is $w_i = (\Delta x)^3$. While straightforward, Cartesian grids waste points in empty regions far from the molecule and lack resolution near nuclei where the density varies rapidly. The error converges polynomially with N.

Becke's partitioning scheme [3] decomposes the molecular integral into atomic contributions:

$$\int f(\mathbf{r})\, d^3 r = \sum_A \int w_A(\mathbf{r})\, f(\mathbf{r})\, d^3 r,$$

where $w_A(\mathbf{r})$ is the Becke weight function centred on atom A, satisfying $\sum_A w_A(\mathbf{r}) = 1$. Each atomic integral is evaluated on a product grid of radial (Gauss–Legendre or Gauss–Chebyshev) and angular (Lebedev) quadrature points, mapped via a radial transform $r = R\,h(x)$ that concentrates points near the nucleus. The Becke radial transform used in this work maps:

$$r = R\,\frac{1+x}{1-x}, \quad x \in [-1, 1],$$

where R is a scaling parameter (set to 1.5 bohr in our implementation). This naturally places more grid points in the chemically important region near the nuclei.

2.1 Method Formulations

The brute-force approach discretises (2) directly:

$$(\mu\nu|\lambda\sigma) \approx \sum_{i=1}^{N} \sum_{j\neq i}^{N} \frac{\rho_\mu\nu(\mathbf{r}_i)\, w_i\; \rho_{\lambda\sigma}(\mathbf{r}_j)\, w_j}{|\mathbf{r}_i - \mathbf{r}_j|},$$

where $\rho_{\mu\nu}(\mathbf{r}) = \chi_\mu(\mathbf{r})\chi_\nu(\mathbf{r})$ is the overlap density. The self-interaction term ($i = j$) is excluded by setting the diagonal of the distance matrix to infinity. This method has $\mathcal{O}(N^2)$ complexity, where N is the number of grid points, making it prohibitively expensive for large grids. Two variants are tested: one using a uniform Cartesian grid and one using the Becke molecular grid.

For a uniform Cartesian grid with spacing Δx, the density is Fourier-transformed via the standard 3D FFT:

$$\widetilde{\rho}(\mathbf{k}) = (\Delta x)^3 \sum_{\mathbf{n}} \rho(\mathbf{r_n})\, e^{-i\mathbf{k}\cdot\mathbf{r_n}}.$$

The ERI is then computed via Parseval's theorem applied to the Coulomb-weighted power spectrum:

$$\mathrm{ERI} = \frac{(\Delta k)^3}{(2\pi)^3} \sum_{\mathbf{k}} \frac{4\pi}{|\mathbf{k}|^2}\, |\widetilde{\rho}(\mathbf{k})|^2,$$

where $\Delta k = 2\pi/(N\Delta x)$. The zero-frequency component ($\mathbf{k} = \mathbf{0}$) is excluded to avoid the Coulomb singularity. The FFT achieves $\mathcal{O}(N^3 \log N)$ complexity for an N^3 grid. However, it requires a uniform grid, preventing adaptive refinement near nuclei.

The one-step NUFFT method uses a Becke grid in real space and transforms directly to either a Cartesian or spherical k-space grid via the type-3 NUFFT:

$$\widetilde{\rho}(\mathbf{k}_j) = \mathrm{NUFFT}_3[\rho(\mathbf{r}_i)\, w_i;\; \mathbf{r}_i \to \mathbf{k}_j]\,.$$

The ERI is then:

$$\mathrm{ERI} = \frac{1}{(2\pi)^3} \sum_j w_j^{(k)}\, \frac{4\pi}{|\mathbf{k}_j|^2}\, |\widetilde{\rho}(\mathbf{k}_j)|^2.$$

For the spherical k-grid variant, the cancellation $w_j^{(k)} \propto k_j^2$ with $1/k_j^2$ simplifies the expression and removes the singularity, as discussed in sec:becke.

This method implements the Poisson equation approach from [4]. Given two overlap densities $f(\mathbf{r})$ and $g(\mathbf{r})$, the algorithm proceeds in four steps:

Algorithm 1 Two-Step Poisson–NUFFT

1: Evaluate $g(\mathbf{r}_i)$ on the real-space Becke grid.
2: **Forward NUFFT:** $\widetilde{G}(\mathbf{k}_j) = \mathrm{NUFFT}_3\big[g(\mathbf{r}_i)\, w_i;\ \mathbf{r}_i \to \mathbf{k}_j\big]$.
3: **Apply Coulomb kernel:** $\widetilde{V}^{(g)}(\mathbf{k}_j) = \frac{4\pi}{|\mathbf{k}_j|^2}\, \widetilde{G}(\mathbf{k}_j)$.
4: **Inverse NUFFT:** $V^{(g)}(\mathbf{r}_i) = \frac{1}{(2\pi)^3}\, \mathrm{NUFFT}_3\big[\widetilde{V}^{(g)}(\mathbf{k}_j)\, w_j^{(k)};\ \mathbf{k}_j \to \mathbf{r}_i\big]$.
5: **Numerical integration:** $\mathrm{ERI} = \sum_i f(\mathbf{r}_i)\, V^{(g)}(\mathbf{r}_i)\, w_i$.

The advantage of this formulation is twofold. First, it decouples the two density distributions f and g, which is essential when $f \neq g$ (i.e., for general four-centre integrals). Second, the intermediate potential $V^{(g)}(\mathbf{r})$ is a smooth function in real space, making the final numerical integration highly accurate.

3 Computational Methodology

3.1 Software Environment

All computations were performed in Python 3.13 using the following libraries:

- **PySCF** [5]: Analytical ERI computation via `mol.intor('int2e')` as the reference baseline.
- **NumPy/SciPy**: Array operations, standard FFT, and Gauss–Legendre quadrature.
- **FINUFFT** [2]: Flatiron Institute Non-Uniform FFT library for type-3 transforms.
- **qc-grids**: Construction of Becke molecular grids and Lebedev angular grids.

3.2 Test System

The benchmark molecule is H_2 with a bond length of 0.74 Å (1.40 bohr), using the `def2-svp` basis set [6]. This basis set was selected for its balance of efficiency and accuracy, and its built-in effective core potential (ECP) support for heavier elements. We compute the simplest ERI, (00|00), where index 0 corresponds to the first GTO basis function. The analytical reference value computed by PySCF is used as the ground truth.

3.3 Density Evaluation

The overlap density for the (00|00) integral reduces to $\rho(\mathbf{r}) = |\phi_0(\mathbf{r})|^2$, where ϕ_0 is the first basis function. This is evaluated using PySCF's `eval_gto` function:

3.4 Grid Construction

Three types of grids are employed, summarised in tab:grids.

Table 1: Grid types used in the benchmark.

Grid	Space	Type
Uniform Cartesian	Real / k	Equally spaced cube $[-L, L]^3$
Becke (molecular)	Real	Atom-centred radial $\times$ angular
Spherical	k	Gauss–Legendre radial $\times$ Lebedev angular

The **Cartesian grid** spans $[-L, L]^3$ with $L = 12$ bohr for real-space methods and $L = 30$ bohr for k-space grids, using N or $2N$ points per dimension. The larger k-space box ensures adequate frequency resolution.

The **Becke grid** is constructed using the `qc-grids` library with N radial Gauss–Legendre points per atom, a Becke radial transform with $R = 1.5$ bohr, and the "fine" angular preset. Points beyond 12 bohr from the origin are clipped to reduce memory usage without sacrificing accuracy, since the GTO density decays exponentially.

The **spherical k-grid** uses N Gauss–Legendre radial points on $[0, k_{\max}]$ with $k_{\max} = 30\ \text{bohr}^{-1}$, and Lebedev angular grids with degree $\ell = \min(\max(2N, 15), 131)$. The angular degree is decoupled from N and capped at 131, the maximum available Lebedev grid order.

The construction of the spherical k-space grid involves mapping Gauss–Legendre nodes from $[-1, 1]$ to $[0, k_{\max}]$:

$$k_i = \frac{k_{\max}}{2}(x_i + 1), \qquad w_i^{(\text{rad})} = w_i^{(\text{GL})}\,\frac{k_{\max}}{2},$$

where $(x_i, w_i^{(\text{GL})})$ are the standard Gauss–Legendre nodes and weights. The full three-dimensional grid is formed as the outer product of radial and angular grids:

$$\mathbf{k}_{ij} = k_i\,\hat{\mathbf{n}}_j, \qquad w_{ij} = w_i^{(\text{rad})}\,w_j^{(\text{ang})}\,k_i^2,$$

where $\hat{\mathbf{n}}_j$ and $w_j^{(\text{ang})}$ are Lebedev unit vectors and weights.

3.5 NUFFT Implementation Details

All NUFFT evaluations use the type-3 transform from the FINUFFT library, which computes:

$$F_j = \sum_{m=1}^{M} c_m\, e^{-i\,\mathbf{k}_j\cdot\mathbf{x}_m}, \quad j = 1, \ldots, N_k,$$

where both the source points $\{\mathbf{x}_m\}$ and target points $\{\mathbf{k}_j\}$ are non-uniform. The tolerance parameter is set to $\varepsilon = 10^{-8}$ for the one-step method and $\varepsilon = 10^{-5}$ for the two-step method. The tighter tolerance in the one-step case is necessary because the entire ERI depends on a single transform, whereas the two-step method benefits from error cancellation between the forward and inverse transforms.

3.6 Benchmarking Protocol

Each method is tested across the grid parameter values $N \in \{2, 4, 8, 12, 16, 32, 64, 130\}$. For each run, we record:

- **Absolute error**: $|\text{ERI}_{\text{numerical}} - \text{ERI}_{\text{exact}}|$,

- **Wall-clock time**: measured via `time.perf_counter()`,
- **Peak memory**: tracked via Python's `tracemalloc` module.

Direct summation methods are capped at $N \leq 32$ to avoid excessive runtime from $\mathcal{O}(N^6)$ scaling (since the grid has N^3 points for Cartesian).

4 Results

fig:benchmark presents the benchmark results across all seven methods as functions of the grid parameter N.

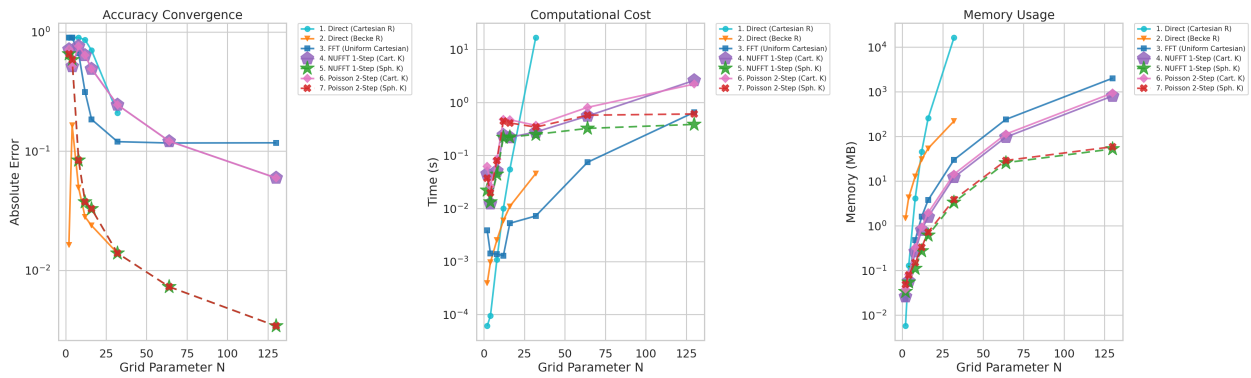

Figure 1: Benchmark results for seven ERI computation methods. **Left:** Absolute error versus grid parameter N (log scale). **Centre:** Wall-clock computation time (log scale). **Right:** Peak memory usage (log scale). Methods: (1) Direct Cartesian, (2) Direct Becke, (3) FFT Uniform, (4) NUFFT 1-step Cartesian k, (5) NUFFT 1-step Spherical k, (6) Poisson 2-step Cartesian k, (7) Poisson 2-step Spherical k.

4.1 Accuracy Convergence

The left panel of fig:benchmark reveals markedly different convergence behaviours:

Direct methods (1–2). The direct Cartesian method (Method 1) converges slowly, with the error plateauing near 10^{-1} even at $N = 32$. This is because the uniform grid poorly resolves the cusp-like density near the nuclei, and the $1/r$ singularity in the Coulomb kernel introduces discretisation artifacts. The direct Becke method (Method 2) converges significantly faster because the atom-centred grid concentrates points where the density is largest. However, both methods share $\mathcal{O}(N^2)$ complexity in the number of grid points, limiting their practical utility.

Uniform FFT (3). The FFT Poisson solver shows moderate convergence, reaching errors around 10^{-1} at large N. Its accuracy is fundamentally limited by the uniform grid: the Nyquist frequency $k_{\max} = \pi/\Delta x$ grows only linearly with N, and the grid cannot adapt to the non-uniform density distribution. The FFT does benefit from excellent computational scaling, however.

NUFFT one-step methods (4–5). Both Cartesian and spherical k-grid variants benefit from the Becke grid in real space, giving them better resolution near the nuclei. The spherical k-grid variant (Method 5, dashed green) converges more smoothly than the Cartesian k-grid

variant (Method 4) because the spherical quadrature is naturally suited to the isotropic Coulomb kernel, and the k^2 cancellation avoids singularity issues. At $N = 64$, Method 5 reaches errors near 10^{-2}.

Poisson two-step methods (6–7). The two-step Poisson solver with spherical k-grid (Method 7, dashed red) has an overlap with the graph of the one-step NUFFT spherical k-grid (Method 5). This is expected since both methods ultimately compute the same integral, but the two-step method benefits from error cancellation between the forward and inverse transforms.

4.2 Computational Cost

The centre panel of fig:benchmark shows wall-clock times on a logarithmic scale.

The direct methods exhibit the steepest scaling, with times reaching nearly one second at $N = 32$ due to $\mathcal{O}(N^6)$ operations for the Cartesian grid. The FFT method scales as $\mathcal{O}(N^3 \log N)$ and is the fastest at small N, benefiting from highly optimised FFTW-based implementations. However, it requires N^3 grid points, so the absolute cost grows rapidly.

The NUFFT methods (4–7) occupy an intermediate regime. Their cost is dominated by the type-3 NUFFT, which scales as $\mathcal{O}((M + N_k) \log(M + N_k))$ where M is the number of real-space points and N_k the number of k-space points. The spherical k-grid variants (5, 7) use fewer k-points than their Cartesian counterparts because the Lebedev grid is more compact in k-space, giving them a consistent speed advantage. At large N, all NUFFT methods maintain sub-second runtimes.

4.3 Memory Usage

The right panel shows peak memory consumption. Direct methods are memory-intensive because they construct the full $N \times N$ distance matrix. The FFT method uses moderate memory for the 3D grid and its transform. The NUFFT methods are the most memory-efficient of the spectral approaches, since they avoid storing dense matrices and operate on sparse, non-uniform point sets. Methods using spherical k-grids consistently use less memory than those with Cartesian k-grids because the spherical grid requires far fewer points to achieve comparable angular resolution.

5 Discussion

As observed in the figures, the superior convergence of Method 7 (Poisson two-step with spherical k-grid) and Method 5 (one-step Nufft with spherical k-grid) can be understood from several mathematical perspectives. However, before we discuss their superiority, we need to acknowledge that both have overlapping convergence rates. This is because mathematically, the two-step Poisson method eventually takes the same form as the one-step method, but with an additional forward and inverse transform. Hence, the convergence rates are expected to be similar, but the two-step method benefits from error cancellation and better conditioning due to the intermediate potential representation, as explained below.

First, the two-step formulation separates the forward transform, kernel application, and inverse transform into distinct stages. This means the NUFFT approximation error enters twice but affects different quantities: the forward transform approximates $\widetilde{G}(\mathbf{k})$ and the

inverse transform reconstructs $V^{(g)}(\mathbf{r})$. Because $V^{(g)}(\mathbf{r})$ is a smooth Coulomb potential (no cusps or singularities away from the nuclei), the inverse NUFFT achieves high accuracy, and the final real-space integration on the Becke grid is exact to high order.

Second, the spherical k-grid exploits the isotropy of the Coulomb kernel. The $4\pi/k^2$ factor depends only on $|\mathbf{k}|$, so a grid with Gauss–Legendre points in the radial direction achieves exponential convergence for smooth radial integrands. In contrast, a Cartesian k-grid wastes points in the corners of the cube where $|\mathbf{k}|$ is large, and the integrand is negligible.

Third, the analytical cancellation of k^2 between the Jacobian and the Coulomb kernel (**??**) and (1) means the effective integrand in k-space is $4\pi\,|\widetilde{G}(\mathbf{k})|^2$ weighted by the base quadrature weights, without any singular factor. This makes the quadrature inherently well-conditioned, unlike the direct methods.

Hence, the direct methods, while conceptually simple, suffer from two compounding issues. The $\mathcal{O}(N^2)$ pairwise summation is inherently expensive, and the discretisation of the $1/r$ singularity introduces systematic errors that diminish slowly with grid refinement. The self-interaction exclusion (setting the diagonal of the distance matrix to infinity) is a necessary but imperfect remedy.

The uniform FFT method is limited by its inability to adapt to the molecular geometry. All points of the density profile, including the near-nuclear cusps, must be resolved by the same uniform spacing. This results in either insufficient resolution (if N is small) or excessive memory and computation (if N is large enough to resolve the cusps).

A consistent finding across both the one-step and two-step methods is the superiority of spherical k-grids. This can be quantified by comparing the number of grid points required: a Cartesian k-grid with N_k points per dimension uses N_k^3 total points, whereas the spherical grid uses approximately $N_{\text{rad}} \times N_{\text{ang}}$ points, where N_{ang} is typically a few hundred (from the Lebedev grid). For the same accuracy, the spherical grid requires one to two orders of magnitude fewer points.

Furthermore, the Cartesian k-grid introduces a regularisation problem at $\mathbf{k} = \mathbf{0}$ (the Coulomb singularity), which is handled by setting $K^2 = \infty$ for small values. This ad hoc treatment introduces a systematic bias, whereas the spherical grid avoids the issue entirely.

The NUFFT tolerance parameter ε plays a critical role. In the one-step method, the entire ERI depends on a single forward NUFFT, so a tight tolerance ($\varepsilon = 10^{-8}$) is required. In the two-step method, the tolerance can be relaxed to $\varepsilon = 10^{-5}$ because errors in the forward and inverse transforms partially cancel when computing the inner product. This tolerance asymmetry is both a practical advantage (faster NUFFT execution) and a theoretical expectation from the variational nature of the energy functional.

6 Conclusion

To reduce the time complexity of complex electron repulsion integrals, which can reach $\mathcal{O}(N^2)$, we must use numerical methods to evaluate them. Overall, these integrals are first constructed analytically using GTO orbitals in PySCF. To calculate these integrals numerically, we have to discretize the integral using quadrature rules. We have used three types of grids: Cartesian, Becke, and spherical. Methods utilizing Fourier transforms (FFT and NUFFT) are used to compute the ERIs in k-space, while direct summation methods are used in real space. The two-step Poisson–NUFFT method is a hybrid approach that

first computes an intermediate potential in k-space and then transforms back to real space for final integration. The two-step Poisson method with a spherical k-grid achieves the best convergence and accuracy due to its ability to handle the Coulomb singularity and to exploit the kernel's isotropy. The one-step NUFFT method with a spherical k-grid also performs well but is limited by the single-transform approximation. Direct methods and uniform FFTs are less accurate and less efficient because they cannot adapt to molecular geometry and handle singularities effectively. Results of the experiment show that both NUFFT techniques show superior convergence and accuracy compared to direct methods and uniform FFT. However, two-step and one-step methods show similar convergence rates, but the two-step method benefits from error cancellation and better conditioning due to the intermediate potential representation.

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